

UGED1111 Logic

Informal Fallacies I

Fallacies in General



A fallacy (謬誤) is a defect in an reasoning that consists in something other than false premises alone.

Two Types of Fallacies

Formal Fallacies

Formal fallacy is a fallacy that can be identified by merely examining the argument form or structure.

Informal Fallacies

Informal fallacy is a fallacy that cannot be identified by merely examining the argument form or structure.

無關論證結構的謬誤

Formal Fallacies (I)

Affirming the consequent (肯定後項)

Given a conditional statement and its consequent,
then its antecedent is deduced.

Deduce p from “If p , then q ” and “ q ”

$$\begin{array}{l} p \supset q \\ q \\ \hline p \end{array}$$

Example:

1. 如果我生病，則我上課睡覺。
2. 我上課睡覺。
3. 我生病。

Formal Fallacies (II)

Denying the Antecedent (否定前項)

Given a conditional statement and the negation of its antecedent, the negation of its consequent is deduced.

Deduce $\sim q$ from “If p , then q ” and “ $\sim p$ ”

$$\begin{array}{l} p \supset q \\ \sim p \\ \hline \sim q \end{array}$$

Example:

1. 如果一日吃十包薯片，則我會變胖。
2. 我沒有一日吃十包薯片。
3. 我不會變胖。

Informal Fallacies



1. Fallacies of Weak Induction
2. Fallacies of Presumption and Ambiguity
3. Fallacies of Relevance

1. Fallacies of Weak Induction

Fallacies of Weak Induction

- ❑ Appeal to Unqualified Authority
- ❑ Hasty Generalization
- ❑ Weak analogy
- ❑ False Cause
- ❑ Slippery slope
- ❑ Appeal to Ignorance

1.1 Appeal to Unqualified Authority

The **appeal to unqualified authority** fallacy occurs in arguments from authority which the cited authority or witness lack credibility.

1. Is the “expert” quoted a recognized expert in the field in question?
2. Is the expert’s assertion based on evidence?

1.1 Appeal to Unqualified Authority

Example:

連大明星米雪都表示某牌子蟲草很有效，吃完少病痛、更健康，所以這個牌子蟲草一定有助健康。



1.2 Hasty Generalization

The fallacy of **hasty generalization** (以偏概全) occurs in an inductive argument when there is a reasonable likelihood that the sample is not representative of the group.

1. The sample size is too small
2. The sample not randomly selected.

1.2 Hasty Generalization



靠北工程師

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「你們知道世界上什麼生肖的人最多嗎？」「龍嗎？」

「不是！是狗！」他很快的公佈解答。

「因為我問班上同學，30個人有25個都是屬狗的！所以世界上屬狗的最多！」

1.3 Weak Analogy



The fallacy of **weak analogy** is committed in an argument which the analogy is not strong enough to support the conclusion.

- The number of relevant similarities
- relevant dissimilarities

1.3 Weak Analogy

Example:

阿樂中大畢業，喜愛哲學，修讀過邏輯，而且擁有一把長髮，他已經有妻子。

阿敏中大畢業，喜愛哲學，修讀過邏輯，而且也擁有一把長髮，所以她應該也有妻子。

1.4 False Cause



The fallacy of **false cause** occurs in an argument whenever the link between its premises and its conclusion depends on some imagined causal connection that probably does not exist.

1.4 False Cause

有时候科学不能解释的事物，也不能用怪力乱神一棍子打死。比如，我前两天画了一道符，家里小强就减少了95.88%！

用的就是普通的蟑螂药笔。

[Translate Tweet](#)



False Cause I

***Post hoc ergo propter hoc* (區前為因)**

After this, therefore on account of this

Example:

我每次考試都先寫上自己的學生編號，結果考試成績都不好，一定是寫學生編號令我成績差！

False Cause II

Non causa pro causa (亂賦因果)

Not the cause for cause

Example

一個好廚師會聽到很多對於其廚藝之讚賞。所以，為了讓我的母親變成好廚師，我決定每天都要稱讚她的廚藝。

False Cause III



Oversimplified cause

Example:

The number of students who receive offers from universities has decreased . Clearly, our teachers just aren't doing their job these days.

1.5 Slippery Slope

The fallacy of **slippery slope** (滑坡謬誤) occurs in an argument which its conclusion rests on an alleged chain reaction that we do not have sufficient reason to believe that it would take place.



1.5 Slippery Slope

Example

如果你這次接受他加價**10%**，那下次你就要接受**20%**，然後他就加**30%**，再來就**40%**，永無止境！

1.6 Appeal to Ignorance

The **appeal to ignorance** fallacy (訴諸無知) occurs when a conclusion is held to be true just because it has not been proved false, or it is false just because it has not been proved true.

1.6 Appeal to Ignorance

Example:

從來沒有人証明過龍不存在，所以龍一定存在。



Discussion 1



我每次看利物浦球賽直播，利物浦都輸掉，
為免他們再輸，我只好不再看直播。

Discussion 2



Discussion 3



你不應該聽你男/女朋友去吃日本餐!這次你遷就他去吃日本餐，下次你得讓他決定你去修哪門課，再下次你又得讓他選擇旅行地點，你以後還能有選擇嗎?

Fallacies of Presumption and Ambiguity

3. Fallacies of Presumption Fallacies of Ambiguity

- ❑ Complex Question
- ❑ Begging the Question
- ❑ False Dichotomy

- ❑ Equivocation

3.1 Complex Question

The fallacy of **complex question** (混合問題) is committed when two (or more) questions are asked in the guise of a single question and a single answer is then given to both of them. Every complex question presuppose a certain condition.

3.1 Complex Question

Example:

女/男朋友: 你有沒有後悔欺騙了我?

3.2 Begging the Question

The fallacy of begging the question (乞求論點) is committed when the arguer supports the conclusion by restating the conclusion as its premise, or by reasoning in a circle.

Example:

1. 我很聰明。
2. 我很聰明。

3.2 Begging the Question

a. Restatement

Example

A: 鬆弛熊是世上最多人喜愛的東西。

B: 你怎知道?好像更多人喜愛熊本熊,HELLO KITTY 呢?

A: 因為根本沒有東西比牠有更多人喜愛。

3.2 Begging the Question

b. Circular Reasoning

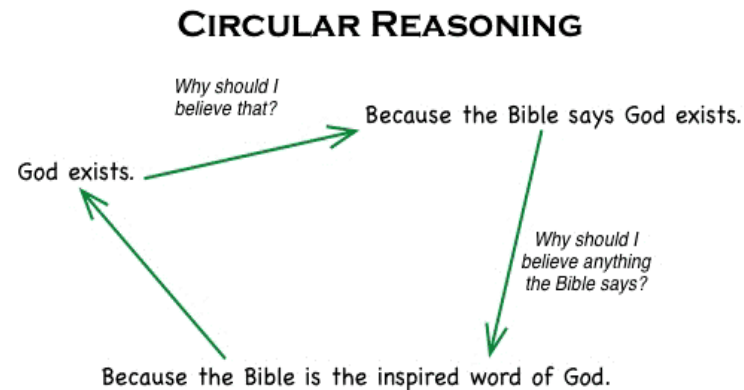
甲：上帝是全能全知全善。

乙：你怎麼知道？

甲：因為聖經是這樣寫的。

乙：那你怎知道聖經寫的東西一定是對的？

甲：因為全能全知全善的上帝所啟示的聖經一定是對的！



3.3 False Dichotomy



3.3 False Dichotomy

The fallacy of **false dichotomy** is committed when two unlikely alternatives is presented in an argument as if they were the only options available, and the undesirable alternative will then be eliminated, leaving the remaining one as the conclusion.

3.3 False Dichotomy

小販:要不你就買光我的所有貨物，要不我就因為一件貨都賣不出去而餓死。作為一個有同情心的人，你不會想我餓死吧？

Fallacies of Presumption and Ambiguity

Fallacies of Presumption

- ❑ Complex Question
- ❑ Begging the Question
- ❑ False Dichotomy

4. Fallacies of Ambiguity

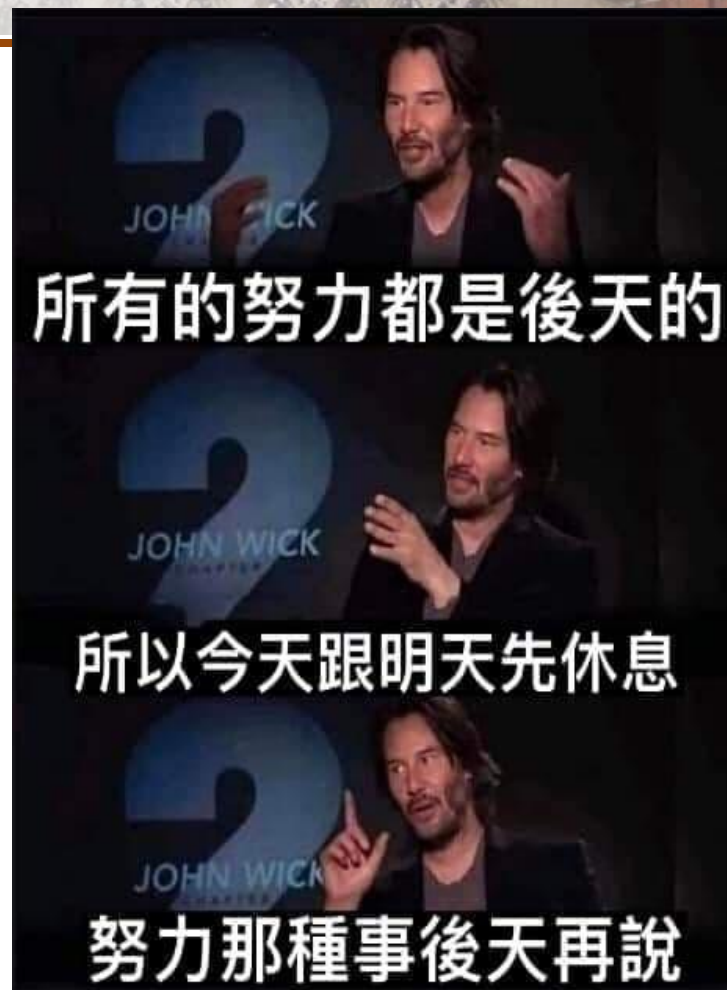
- ❑ Equivocation

4. Equivocation



The fallacy of **equivocation** is committed when the conclusion of an argument depends on the fact that a word or phrase is used, either explicitly or implicitly, in two different senses in the argument.

4. Equivocation



4. Equivocation



Example

Noisy children are a real headache. Two aspirin will make a headache go away. Therefore, two aspirin will make noisy children go away.

Discussion 4



小美: 小波是個誠實人。

小儀: 真的嗎? 我好像... 記得...

小美: 他自己說自己是不會說謊的。

小儀: 他說就是真話嗎?

小美: 既然他是誠實人，他當然不會欺騙我!

Discussion 5



當杯子塔最高的杯裝的水多了滿了，水自然會流向下層其他杯子。同樣地，當社會上層的人錢多了，我們可以預計錢也會被下層的人分享。

Discussion 6



通過電郵，調查訪問了**5000**位市民，題目為「你有使用電郵嗎？」，最後發現**100%**的受訪者都使用電郵。調查反映所有人都有使用電郵

Discussion 7



要不你就接受用米埔濕地公園來興建房屋，要不你就接受香港沒有足夠房屋，選擇顯然易見吧。

Discussion 8



每當母親提醒我出門要帶雨傘，那天總是會下大雨，下大雨外出會非常麻煩，為免我出門時遇上大雨，我決定阻止母親再對我作出提醒。

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